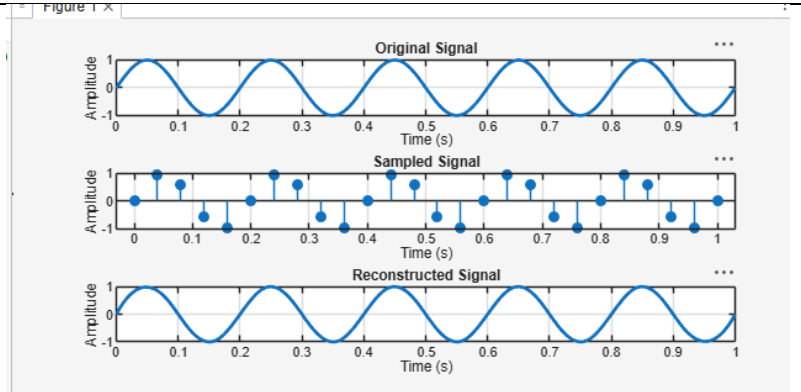


7. SIGNAL RECONSTRUCTION AND SPECTRAL ANALYSIS

Experiment 7.1

Reconstruction of Signal from Samples (Ideal Interpolation)

```
% Reconstruction of Signal  
from Samples  
clc;  
clear;  
close all;  
% Original signal frequency  
fm = 5; % Hz  
% Sampling frequency  
fs = 25; % Hz  
% Continuous-time signal  
t = 0:0.001:1;  
x = sin(2*pi*fm*t);  
  
% Sampling  
ts = 0:1/fs:1;  
xs = sin(2*pi*fm*ts);  
% Reconstruction using  
interpolation  
xr =  
interp1(ts,xs,t,'spline');  
% Plotting  
figure;  
subplot(3,1,1);  
plot(t,x,'LineWidth',2);  
grid on;  
title('Original Signal');  
xlabel('Time (s)');  
ylabel('Amplitude');  
subplot(3,1,2);  
stem(ts,xs,'filled');  
grid on;  
title('Sampled Signal');  
xlabel('Time (s)');  
ylabel('Amplitude');  
subplot(3,1,3);  
plot(t,xr,'LineWidth',2);  
grid on;  
title('Reconstructed  
Signal');  
xlabel('Time (s)');  
ylabel('Amplitude');
```



Experiment 7.2

Fourier Series of Half-Wave Rectifier Output

```
% Fourier Series of Half-Wave  
Rectifier Output  
clc;  
clear;  
close all;  
% Time axis  
t = 0:0.001:2*pi;  
% Half-wave rectified signal  
x = sin(t);  
x(x < 0) = 0;  
% Fourier Series approximation  
N = 10; % Number of harmonics  
a0 = (1/pi)*trapz(t,x);  
fs_signal = a0/2;  
for n = 1:N  
    an =  
(1/pi)*trapz(t,x.*cos(n*t));  
    bn =  
(1/pi)*trapz(t,x.*sin(n*t));  
  
    fs_signal = fs_signal +  
an*cos(n*t) + bn*sin(n*t);  
end  
% Plotting  
figure;  
subplot(2,1,1);  
plot(t,x,'LineWidth',2);  
grid on;  
title('Half-Wave Rectified  
Signal');  
xlabel('Time');  
ylabel('Amplitude');  
subplot(2,1,2);  
plot(t,fs_signal,'LineWidth',2)  
;  
grid on;  
title('Fourier Series  
Approximation');  
xlabel('Time');  
ylabel('Amplitude');
```

